Marker length: what the results show

FYP: synchronization markers for DNAFormer · fine-tuned models, test sets of 1,000 strands per deletion level · 29 Oct 2026
Runs analysed: no markers; periodic marker lengths 4, 5, 6, 7, 8 and 12; and the length-8 end-block control.

38.8% → 62.3%
perfect strands at 8% deletion: no markers → 12-letter markers

4.9% → 1.9%
wrong letters left after decoding at 8% (cut by 60%)

≈ 50 letters
data recovered per strand at 8% for every marker length, 0 to 12 (no net gain)

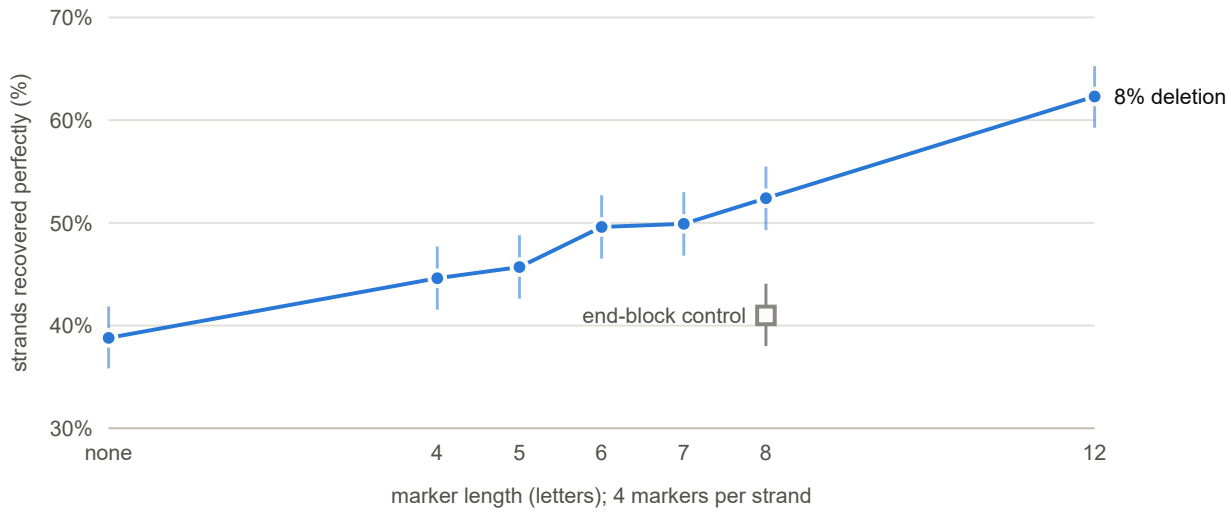
Key insights

1. **Longer markers give more reliable strands, all the way up to 12 letters.** At 8% deletion, perfect recovery rises with marker length: none 38.8%, 4 → 44.6%, 5 → 45.7%, 6 → 49.6%, 7 → 49.9%, 8 → 52.4%, 12 → 62.3%. Wrong letters after decoding fall from 4.9% to 1.9%. Length 12 is clearly better than 8 (+9.9 points, $p < 0.001$), so **no peak has been reached** in the range tested.
2. **The gain comes from placement, not just from having fewer data letters.** With the same 8-letter markers placed together at the end (end-block control), recovery is only 41.0%, against 52.4% when they are spread out ($p < 0.001$). So the markers carry real synchronization information.
3. **Markers trade data space for reliability almost exactly one-for-one.** At 8% deletion every version, from no markers to 12-letter markers, recovers **49–52 data letters per strand**. Each extra marker letter removes data space at the same rate as it adds reliability. At 5% deletion, no markers recovers the most data (120 vs 78 at length 12).
4. **So there is no single "best" marker length. It depends on what you optimize.** For reliability per strand, longer is better (12 is best of those tested). For raw data recovered, all lengths tie. For information after an ideal error-correcting code, shorter is better (no markers is highest; see Figure 4).
5. **At 1% and 3% deletion, markers do not matter.** Every version recovers 99.5–100% of strands perfectly. Markers only help when deletions are heavy, and clearly only at 8%.

> **Bottom line.** Periodic markers are real synchronization aids: they make each strand far more reliable under heavy deletions, and more marker letters means more reliability (up to 12 tested). But they do not increase how much data a strand delivers. Their value depends on the error-correcting code used on top, which is your supervisor's question and the next step.

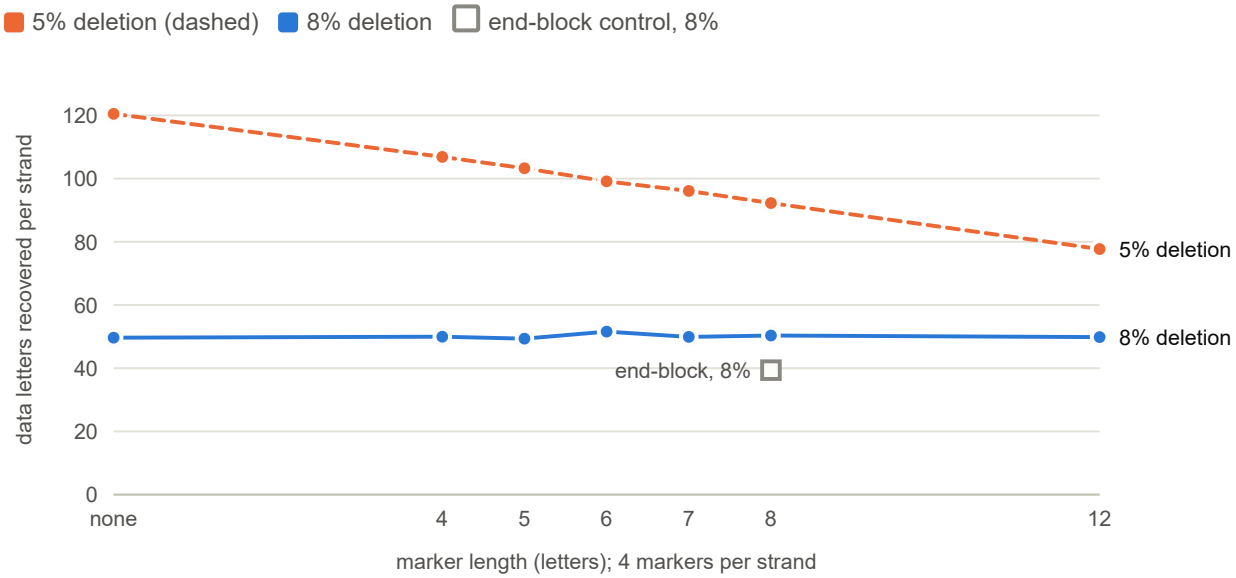
Figures

1. Perfect strand recovery at 8% deletion



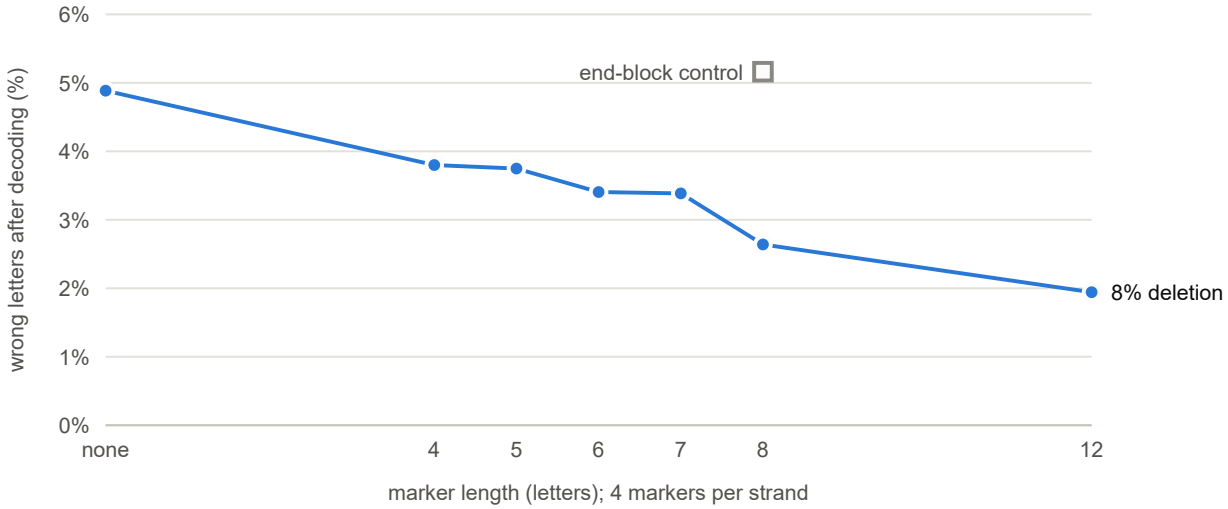
Dots: share of 1,000 test strands recovered with zero errors (data letters only). Bars: 95% confidence intervals. Gray square: 8-letter markers placed together at the end of the strand (control). Recovery keeps growing with marker length up to 12; spreading the markers out is what helps.

2. Data letters recovered per strand



Perfect strands × data letters per strand. At 8% the line is flat (≈50 letters for every length from 0 to 12): what markers gain in reliability they lose in space. At 5% markers lose data, because almost all strands are recovered anyway.

3. Wrong letters left after decoding (8% deletion)



Share of data letters that are still wrong after DNAFormer (1 − per-letter accuracy). Longer markers cut the error rate by more than half (4.9% → 1.9%). Placed at the end, the same markers make it slightly worse than no markers.

4. How much information an ideal error-correcting code could deliver (8% deletion)

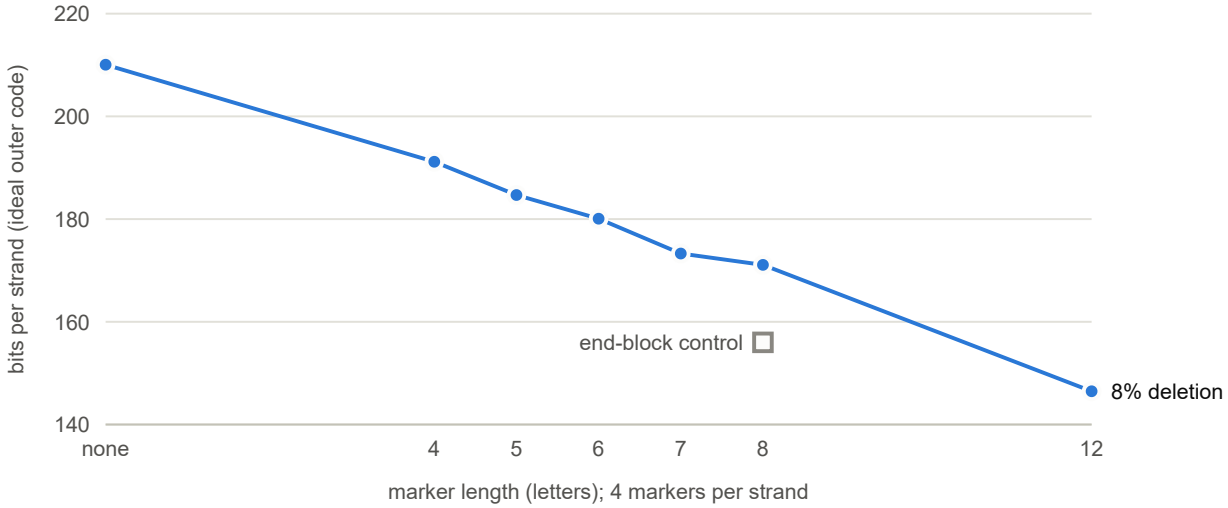


Illustration only: bits per strand = data letters × (2 − H), where H is the information lost to the remaining letter errors, as if an ideal outer code fixed them. It treats leftover errors as independent letter swaps, which is optimistic. Under this estimate no markers still delivers the most information (210 vs 171 bits), so markers only pay off if they make a real code work much better, which has to be tested.

Full results

Version	Marker	Data letters	Marker share	5%: perfect	8%: perfect [95% CI]	8%: NED	8%: wrong letters	8%: data / strand
marker_free	-	128	0.0%	94.1%	38.8% [35.8, 41.9]	0.0202	4.89%	49.7
marker_len04	`CAGT`	112	12.5%	95.4%	44.6% [41.5, 47.7]	0.0183	3.80%	50.0
marker_len05	`TCATG`	108	15.6%	95.6%	45.7% [42.6, 48.8]	0.0182	3.75%	49.4
marker_len06	`CTGTCA`	104	18.8%	95.3%	49.6% [46.5, 52.7]	0.0172	3.41%	51.6
marker_len07	`CTGATCA`	100	21.9%	96.1%	49.9% [46.8, 53.0]	0.0168	3.39%	49.9
marker_periodic	`ACGTCAGT`	96	25.0%	96.1%	52.4% [49.3, 55.5]	0.0148	2.64%	50.3
marker_len12	`ACGTCAGTGCAT`	80	37.5%	97.1%	62.3% [59.3, 65.3]	0.0118	1.94%	49.8
marker_endblock	`ACGTCAGT (end)`	96	25.0%	95.7%	41.0% [38.0, 44.1]	0.0241	5.16%	39.4

Perfect = strand recovered with zero errors. NED = normalized edit distance (lower is better). Wrong letters = 1 − per-letter accuracy. Data/strand = perfect × data letters. Highlighted: the 8-letter periodic run from Stage 4. At 1% all versions are 100%; at 3%, 99.5–100%. File-level recovery (25-strand files with no error correction) is 0 of 40 for every version at 8%.

Is the difference real? (8% deletion)

Comparison at 8% deletion	Difference	z	p (approx.)	Verdict
Length 8: spaced vs end-block (synchronization)	+11.4 points	5.1	<0.001	clear
Length 4 vs no markers	+5.8 points	2.6	0.008	clear
Length 6 vs length 4	+5.0 points	2.2	0.025	weak
Length 8 vs length 6	+2.8 points	1.3	0.210	not significant
Length 8 vs length 7	+2.5 points	1.1	0.263	not significant
Length 8 vs no markers	+13.6 points	6.2	<0.001	clear
Length 12 vs length 8	+9.9 points	4.5	<0.001	clear
Length 12 vs no markers	+23.5 points	10.8	<0.001	clear

Two-proportion z-test on perfect recovery (1,000 strands each). This ignores the pairing (same strands, same errors), so it is conservative: a paired McNemar test on predictions.csv will be at least as strong. Run that for the final report.

Caveats

- **One training run per version (seed 1).** Differences of 1–3 points (for example 6 vs 7 vs 8) can come from training luck. Retrain the top 2–3 lengths with a second seed before claiming a single best.
- **Simulated data only**, and at deletion rates (5–8%) far above real nanopore reads (~1%).

- **Only one sequence per length.** We showed earlier that sequence matters little for random data, but it was not tested by training.
- **Exact recovery is strict.** One wrong letter counts as a failure, which is why "data recovered" and "wrong letters" are also shown.

Recommended next steps

1. **Find where reliability stops improving:** try 16-letter markers (64 marker letters, 50%) to see whether the curve flattens.
2. **Run paired tests** (McNemar on predictions.csv) for 12 vs 8, 8 vs 7, and 8 vs no markers.
3. **Second seed** for lengths 6, 7, 8 and 12.
4. **Add an outer error-correcting code** (for example Reed–Solomon across strands) and compare: (a) no markers plus more parity, against (b) markers plus less parity, at the *same total overhead*. This is the fair test of whether markers are worth their cost, and it answers your supervisor's question directly.
5. **Optionally, test 10% deletion**, where markers should help more.

> **What you can claim now:** "Periodic synchronization markers significantly improve DNAFormer's per-strand reliability under heavy deletions (8%), and the benefit keeps growing with marker length up to the longest tested (12 letters); the improvement comes from marker placement. However, the reliability gained matches the storage the markers use almost exactly (about 50 data letters recovered per strand at every length), so their net value depends on the outer error-correcting code."